Computer Organization and Architecture	
Lab #:	**5**
Lab Title:	Display the result of computing the expression & Change the case of the character from lower to upper and from upper to lower and display it.
Submitted by:	

Names	**Registration #**
Sumaira Safeer	FA22-BCE-019

Rubrics name & number					**Marks**	
					In-Lab	**Post-Lab**
Engineering Knowledge	***R2: Use of Engineering Knowledge and follow Experiment Procedures:*** *Ability to follow experimental procedures, control variables, and record procedural steps on lab report.*					
Problem Analysis	***R6: Experimental Data Analysis:*** *Ability to interpret findings, compare them to values in the literature, identify weaknesses and limitations.*					
Design	***R8: Best Coding Standards:*** *Ability to follow the coding standards and programming practices.*					
Modern Tools Usage	***R9: Understand Tools:*** *Ability to describe and explain the principles behind and applicability of engineering tools.*					
Individual and Teamwork	***R12: Individual Work Contributions:*** *Ability to carry out individual responsibilities.*					
	R13: Management of Team Work: *Ability to appreciate, understand and work with multidisciplinary team members.*					
Rubrics #	**R2**	**R6**	**R8**	**R9**	**R12**	**R13**
In Lab						
Post- Lab						

LAB: 05

Task 1:

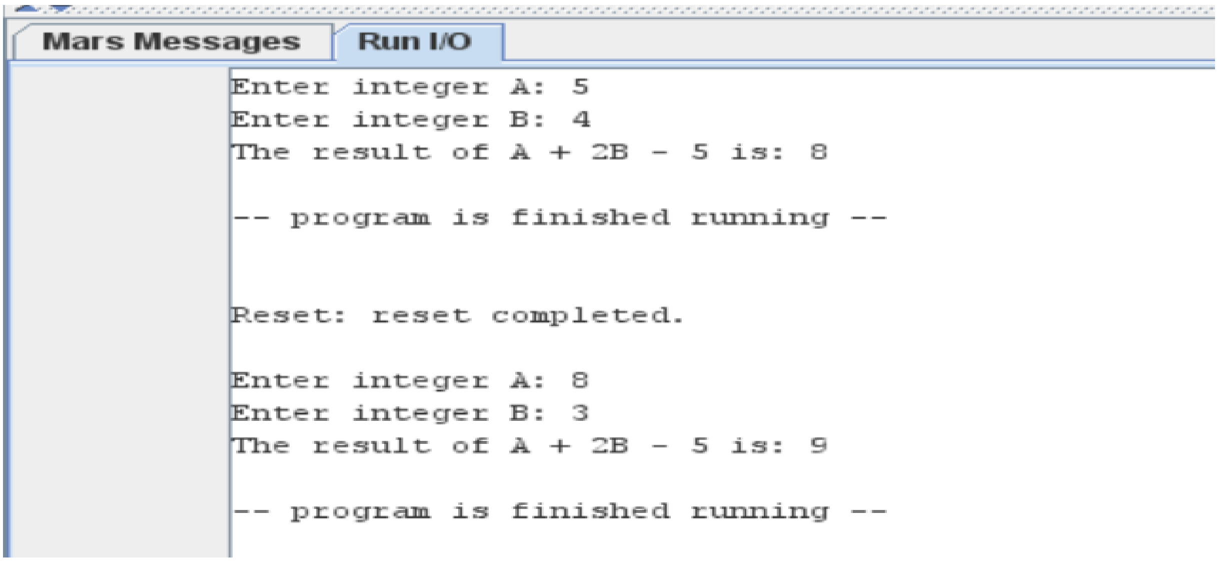
Write a program to ask the user to enter two integers A and B and then display the result of computing the expression: A + 2B - 5.

Code:

```
.data
prompt_a:       .asciiz "Enter  integer A: "   # Prompt for A
prompt_b:       .asciiz "Enter integer B: "   # Prompt for B
result_msg:     .asciiz "The result of A + 2B - 5 is: "
newline:       .asciiz "\n"
.text
.globl main

main:
    li $v0, 4                    #  print string
    la $a0, prompt_a              # load address of prompt_a
    syscall
    li $v0, 5
    syscall
    move $t0, $v0                # store A in $t0

    li $v0, 4
    la $a0, prompt_b              # load address of prompt_b
    syscall
    li $v0, 5
    syscall
    move $t1, $v0                # store B in $t1

    # Compute A + 2B - 5
```

```
add $t2, $t1, $t1            # $t2 = 2B (B + B)
add $t3, $t0, $t2            # $t3 = A + 2B
sub $t4, $t3, 5              # $t4 = A + 2B - 5

li $v0, 4
la $a0, result_msg           # load address of result_msg
syscall

# Print the result
li $v0, 1
move $a0, $t4                # load result into $a0
syscall
li $v0, 4
la $a0, newline              # load address of newline
syscall
li $v0, 10                   # exit
syscall
```

Output:



Task 2:

Write a program that asks the user to enter an alphabetic character (either lower or upper case) and change the case of the character from lower to upper and from upper to lower and display it.

Code:

```
.data
    prompt:     .asciiz "Enter an Alphabetic Character: "  # Input prompt
    result:     .asciiz "The character in the Opposite Case is: "
    invalid:    .asciiz "Invalid input. Please enter an alphabetic character.\n"
    newline:    .asciiz "\n"
.text
.globl main

main:
    li $v0, 4                  # syscall to print string
    la $a0, prompt             # load the prompt message
    syscall
    li $v0, 12                 # syscall to read a character
    syscall
    move $t0, $v0

    # Check if character is lowercase ('a' to 'z')
    li $t1, 97                 # ASCII value of 'a'
    li $t2, 122                # ASCII value of 'z'
    blt $t0, $t1, check_upper  # If less than 'a', check uppercase
    bgt $t0, $t2, check_upper  # If greater than 'z', check uppercase

    li $t3, 32                 # ASCII offset between cases
    sub $t0, $t0, $t3          # Convert to uppercase
```

```
    j display_result          # Jump to display result

check_upper:
    # Check if character is uppercase ('A' to 'Z')
    li $t1, 65                # ASCII value of 'A'
    li $t2, 90                # ASCII value of 'Z'
    blt $t0, $t1, invalid_input   # If less than 'A', invalid input
    bgt $t0, $t2, invalid_input   # If greater than 'Z', invalid input
    li $t3, 32
    add $t0, $t0, $t3         # Convert to lowercase

display_result:
    li $v0, 4
    la $a0, result            # load the result message
    syscall
    li $v0, 11                # syscall to print a character
    move $a0, $t0             # load the converted character
    syscall
    li $v0, 4
    la $a0, newline           # load the newline
    syscall
    li $v0, 10                # syscall to exit
    syscall

invalid_input:
    li $v0, 4
    la $a0, invalid           # load the invalid input message
    syscall
    li $v0, 10                # syscall to exit
    syscall
```

Output:

ages | Run I/O

```
Enter an Alphabetic Character: SThe character in the Opposite Case is: s

-- program is finished running --


Reset: reset completed.

Enter an Alphabetic Character: aThe character in the Opposite Case is: A

-- program is finished running --


Reset: reset completed.

Enter an Alphabetic Character: 6Invalid input. Please enter an alphabetic character.

-- program is finished running --
```